Contact

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Skills

Generative AI: Prompt Engineering, Hugging Face, Transformers
Deployment, Containerisation
Docker, TailScale, SkyPilot, Flask, Streamlit,
DBMS:
MySQL, PostgreSQL, MongoDB, MariaDB
Analytics & Visualisation:
Python, Power BI, MS-Excel
Data Science Libraries:
Pytorch, Keras, sklearn, pandas,
Programming: Data Versioning, HTML,CSS,

Certifications

Google IT Support Certification,
Introduction to Linux,
SQL for Data Science,
Machine Learning Foundations: A Case Study Approach ,
Crash Course on Python

Teaching

Teaching Assistant, NPTEL
(Jan 2024- April 2024)
noc24-cs47:

The Joy of Computing Using Python
Prepared weekly assignments & handled the forum for the course having around 60,000 registered learners.

Student Mentor, IITM BS Degree Programme
(Mar 2024- April 2024)
BSMA1004:

Statistics for Data Science 2.
Conducted doubt clearing sessions of 2 hours, 3 times a week for a set of 25 students

Volunteering

COVID-19 Response Volunteer:
Caregiver Saathi, May 2021

Maria Thurkadayil

DATA SCIENTIST

Summary

Master's Degree Student building skills in the field of Data Science, Prior background in Computer Science and Engineering. Looking for opportunities to work with latest technologies

Experience

Graduate Research Intern , IIT Jammu | June 2026- Present
https://iitjammu.ac.in/mira/
Gaperon mmbert distill repository (Ongoing)
- Worked on a project using jhu-clsp/mmBERT-base with 6 independent classification heads to predict text quality (clarity, coherence, depth, grammar, usefulness, overall) across English, Hindi, and Tamil. Enhanced the model to include datasets for languages like Bengali, Telugu and Malayalam
- Set up end-to-end pipeline using SkyPilot with HuggingFace Hub integration for dataset streaming, checkpoint versioning, and model promotion across branches

Synthia - Synthetic Data Generation Pipeline (Ongoing)
- Worked on a Python pipeline for generating instruction fine-tuning data (context, instruction, response triples) from raw text documents using self-hosted LLMs
- Set up and validated the local development environment, connecting to MIRA's private GPU inference cluster over Tailscale VPN.

Summer Intern, S3 Labs, IIT Bhiliai | May 2025- July 2025
1. **Technical Image Understanding**
   - Evaluating how Packet Diagrams, Sequence Diagrams, Class Diagrams, State Diagrams, etc are inferenced using LLM's. using a set of hand drawn images and mermaid code diagrams
2. **Legal Infographic Generation** Dataset : IL-TUR(Exploration Lab, IIT Kanpur)
   - Conducted Research Paper Analysis on the Applicability of LLM's in the Legal Domain
   - Generated trial prototypes for generating Infographics like Timelines, Key Arguments

Supervisor: Gagan Raj Gupta, Department of Computer Science and Engineering

Projects

1. **Forecasting the Total Marine Fish Production In Kerala - Development & Comparative Analysis of Machine Learning Models**
   Machine Learning, Fisheries | Jan 2022- May 2022
   Technology : SPSS, scikit-learn, pandas
   - Marine Fish production Data in Kerala (in tonnes) was collected from **CMFRI, Kochi** and data on 9 significant environmental variables from **ICOADS** from 1960 - 2014 .
   - Highest predictive power - Random Forest- R2 score 0.87, MAE 0.05, MSE 0.005, RMSE 0.07
2. **Chat Application- Publisher Subscriber Architecture | November 2019**
   - Developed a real-time communication system using the MQTT protocol, applying a publisher–subscriber model to support chat-style applications as part of Hack-Kochi#2
3. **VR Industrial Training Simulator | Unity, C# | Jan 2021**
   Co-developed a Virtual Reality application simulating boiler feedwater plant control room operations, creating a safe, immersive environment for industrial operator training
   Collaborated on engineering interactive 3D scenarios and automation workflows in Unity/C#, effectively eliminating the physical risks and costs associated with traditional models.
4. . **Prediction of Crime Categories**
   Ensembled Learning, Law & Order | June 2024–August 2024
   Technology: xgboost, lightgbm, pandas, scikit-learn
   - The dataset consisted of 20,000 rows & 20 columns about location, victim

Others

.GATE DA 2025: Rank 3876
(March 2025)

Winner Hack Kochi #2:
(November 2019)
IEEE ORE, Kochi Hub

KEAM- Engineering: Rank 333
(August 2018)
Kerala State Entrance Examination

Highest Scorer: Science Stream:
Subject Topper for Maths, Chemistry and Computer Science
(May 2018)

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demographics, date, time, etc. 6 crime categories were possible for the target variable.

- Data Exploration, Cleaning and Preprocessing created 23 relevant feature columns.
- **LGBMClassifier()** and **XGBoostClassifier()** were used in building a **VotingClassifier()** for which had an accuracy of 0.88080 in contrast to the baseline accuracy of 0.58

.5. **Perspective Aware Sentiment Analysis**
Applied Data Analytics, Sentiment Analysis | November 2025
Models: Roberta-base, GPT-4o-mini
Technology: transformers, openai, feedparser, pandas, python-dotenv

- Real-time data pipeline using **Google News RSS** to fetch and parse headlines based on specific search queries and time windows.
- Implemented a standard sentiment baseline using the **cardiff nlp/twitter-roberta-base**-sentiment-latest model to provide objective scores. Perspective-aware sentiment engine was created using GPT-4o-mini that adapts its analysis based on user-defined personas .
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5. **Differential Privacy in PyTorch (Opacus)**
Open Source Contribution, Data Privacy and Data Quality Project | May 2026

- Attempted a Contribution to Opacus, Meta's open-source library enabling differentially private deep learning with minimal code changes.
- Worked on an issue that had a privacy bug (silent epsilon miscalculations with WeightedRandomSampler)

6.. **LLM Distillation for Named Entity Recognition**
Applied Machine Learning Project | May 2026
Collaborated with a teammate to develop a knowledge distillation pipeline, compressing a 12B-parameter LLM (Gemma 3) into a 270M-parameter model for fine-grained NER.
Helped validate the final model, which achieved a strict-match F1 score of 0.698, successfully matching the teacher model's performance at a fraction of the compute cost.

7 . **Dynamic Reflections: Probing Video Representations with Text Alignment |** Introduction to Generative AI| June 2026

- Partnered with a teammate to evaluate and present cutting-edge research on multimodal video-text representation alignment and spatio-temporal semantics for the "Introduction to Generative Intelligence" course at CMI.
- Explored accompanying Jupyter Notebook implementations of zero-shot probing techniques and analyzed cross-modal semantic alignment methodologies, culminating in a comprehensive technical presentation of the findings

Education.

Chennai Mathematical Institute,
Data Science, Aug 2024 - May 2026

Master of Science
CGPA : 7,16

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